

what
is
slop-
ware
,
how
do
you
spot
it,

×

what should I call you?

ok

this name stays in this browser. it is not sent anywhere.

riculum
that
teaches
you to
measure
the dis-
tance
between
some-
thing
that
looks
like it
works
and
some-
thing
that ac-

tually
holds
up.
every
chapter
ends
with a
check
you run
on your
own
project.

6 / 55
chapters
· pdf ·
every

what should I call you?

this name stays in this browser. it is not
sent anywhere.

00 why
does
this
book
exist?
01 what
is the
thing
on your
screen?
02 why
can
your

deploy
not
find
your
files?

03 why
does
your
link
only
open
for
you?

04
what

what should I call you?

this name stays in this browser. it is not
sent anywhere.

npm
run
dev
and
npm
run
build?

05 why
does
the
till
stand
in a
dif-

ferent
room?

what should I call you?

this name stays in this browser. it is not
sent anywhere.

00

why does this book ex- ist?

Why this
book ex-
ists,
who is
writing
it, and
how to

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Reddit,
Quora and
Ekşi
Sözlük for
years, and
once I was
one of the
people
posting
there.
People
make fun
of local-
host:3000
users, and
they be-

lieve CS is
dead be-
cause
somebody
developed
a local
website.
Actually, I
hate need-
less jokes
made just
to fit in.
But also,
lack of
technical
depth gave
those lo-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

joining in,
I decided
to create a
blog about
how to
avoid slop-
ware,
based on
my own
experi-
ences.

Learnin
g it cost
me 53 re-
pos, 8 pull
requests
and 6

Claude
Max x20
accounts.
It also
cost me
exploding
token bills,
sleepless
nights,
and the 5
kilos I lost
to hunger
and lack
of sleep. I
do not
want that
to happen

what should I call you?

this name stays in this browser. it is not
sent anywhere.

den inside
one class
of people.
I am using
Claude
while I
write this,
and I am
sure Dario
Amodei
would
want the
same. So I
am writing
it with my
own
hands, out

of every-
thing I
pulled
from my
own pain.

I do
not believe
that CS
has been
replaced
by AI.*
Over time
there was
COBOL
and C++
did not re-
place it,

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Java.

Given the
nature of
high level
languages,
I treat an
LLM as
one more
high level
language.

(And
where has
engineer-
ing gone?
Engineerin
g is the

difference
in the
thinking
itself, in
knowing
what to
make and
in con-
necting
technolo-
gies and
developing
them.)

I love
vibecod-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

limited to-
kens. As
far as I
see, in-
ternships
and
projects
are pre-
sented as
though
they be-
long to
certain
people.
What gets
inherited
is not only

money.

You can
call it in-
herited vi-
sion, you
can call it
social
climbing,
and there
is even a
niche for it
now,
something
like a class
based net-
working
hub

what should I call you?

this name stays in this browser. it is not
sent anywhere.

business
with a
Claude
Pro ac-
count and
has gradu-
ates of
good
schools
working
under-
neath
them,
when
those same
students
could have

started
that com-
pany with
their
pocket
money.
Neverthele
ss, if it is
not rocket
science
and it is
not tank
design, I
do not
think any
work that
needs no

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Your ap-
plication
was reject-
ed, so say
that you
could do
this too
and carry
on your
way.

Everyo
ne should
benefit
from tech-
nology,
and not
only for

business
and mon-
ey, but for
making
products
for your
own needs.
As I be-
lieve from
Plato, no
one does
wrong
willingly,
and no-
body
writes bad
software

what should I call you?

this name stays in this browser. it is not
sent anywhere.

different
from being
able to
sew a but-
ton on.
Everyone
should
learn it in
order to
make the
ideas they
dream of
real, and
most im-
portantly,
everyone
can, be-

cause I be-
lieve that
for once in
history
anyone
can build
anything.

What
matters is
whether
anything
stands be-
hind the
code.

Think
about the
last time

what should I call you?

this name stays in this browser. it is not
sent anywhere.

tests ran
and none
of them
com-
plained,
and noth-
ing about
whether
they
would
have no-
ticed a
broken
app.

Who am
I?

I am
Damla, a
CS stu-
dent at
Bilkent. I
was on the
board of
IEEE, I
worked as
a Python
assistant, I
did an in-
ternship
and a fel-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

fore CS
took over,
starting
with
HTML at
11 and a
lot of
Stack
Overflow.
Now I
vibecode
anyway.
(Okay,
what I re-
ally do is
LLM sys-

tems engi-
neering,
which
means I
connect
systems
and engi-
neer
them.)

ir-globe
draws
what 198
countries
do to each
other and
it has
been going

what should I call you?

this name stays in this browser. it is not
sent anywhere.

costs 0
dollars a
month.
stitchu
takes a
photo-
graph of a
garment
and gives
back a
sewing
pattern,
and it has
a gate
that de-
cides
whether

that pat-
tern can
actually
be sewn,
which I
wrote af-
ter print-
ing pat-
terns that
passed
every test
I had and
could not
be sewn at
all. lu-
lumelon
measures

what should I call you?

this name stays in this browser. it is not
sent anywhere.

rather
than a sin-
gle num-
ber, be-
cause on
its first
live run
the honest
interval
ran from 0
to 33,3.
rabadon is
a gate sys-
tem for
agents,
written in
C++ with

no dependencies,
and it exists because an agent once told me it had finished a job it had not finished.

Each of those started as something I got wrong,

what should I call you?

this name stays in this browser. it is not sent anywhere.

over.

How should you read this book?

The order is semantic, and I spent a full week reading and think-

ing before
I settled
it. I start-
ed from
localhost,
because
that is the
joke I kept
seeing.
Writing it
made me
realise the
link only
fails if you
believe a
page is a
place, so

what should I call you?

this name stays in this browser. it is not
sent anywhere.

was talk-
ing about
files with-
out ever
saying
what a
folder is,
so the ter-
minal
came next.

Localho
st then re-
quired
npm run
dev, be-
cause once
you know

whose machine answers you want to know what is running on it. That one required npm run build, because the thing you would put on another machine is not the

what should I call you?

this name stays in this browser. it is not sent anywhere.

program on another machine is what the whole book has been walking towards. The backend required data, because a backend keeps something.

Data re-
quired
.env files
and API
keys, be-
cause the
moment
something
is kept,
whatever
reaches it
matters.

Those
chapters
are
grouped
into five

what should I call you?

this name stays in this browser. it is not
sent anywhere.

project off
your own
computer.

301 is
where real
users and
real at-
tackers ar-
rive, 401 is
the part
people
touch, and
501 is
money.

Afterward
s I will
write

about
LLM sys-
tem de-
sign, or-
chestra-
tion,
loops,
workflows
and gates.

Every
chapter
opens with
a real inci-
dent and
closes with
the thing
you can do

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ule be-
cause 2
months in
tech = 10
years of
develop-
ment in
any topic.

By the
way, new
tabs keep
opening in
my head,
and a
chapter
about one
thing

wakes up
an idea
about
something
else entirely.
Those
ideas do
not belong
inside the
chapter, so
they go in
the appendix,
and terms
and small
technical
things

what should I call you?

this name stays in this browser. it is not
sent anywhere.

tom of the
page, so
anything
marked
with a star
has a note
waiting for
it.

The appendix is
also where
the writing about
product
management and
the start-

up world
will go.
Companies
still run
events to
hand out
the kind of
knowledge
you can
only get
through
experi-
ence, and
then they
turn cre-
ative peo-
ple away

what should I call you?

this name stays in this browser. it is not
sent anywhere.

were elimi-
nated. To
tell you
the truth,
I do not
think any-
body is as
well inten-
tioned as I
am, and
people
hold a
strange
principle
about not
sharing
good

things.

You can
steal my
ideas.

There is
sand in
the sea
and there
are ideas
in me, so
passing on
what I
know is
nothing
but a hap-
piness for
me.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

tation
made me
interested
in comput-
er science,
and I
changed
my de-
partment.
Even
though we
have not
been writ-
ing code
by hand
for the

last year,
it gave me
a love of
CS that
no under-
graduate
degree
could have
given me.
We used
to shorten
loops with
algorithm
analysis,
and now
we do the
same thing

what should I call you?

this name stays in this browser. it is not
sent anywhere.

changed,
while the
purpose
and the
logic did
not, so I
still think
this mo-
tion in CS
is a matter
of
perspec-
tive.

So
when I am
no longer
a sweet

student
but a busi-
ness-
woman, I
hope this
book will
have en-
lightened
somebody,
enter-
tained
somebody,
or made
somebody
think.

Two
heads are

what should I call you?

this name stays in this browser. it is not
sent anywhere.

too. My
endless re-
spect to
everyone
who is
dreaming
and work-
ing for
something.

* Is CS
dead?
The
claim

is
usual-
ly
made
about
a lan-
guage
, not
about
the
field.
COB
OL
was
going
to be

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Java
by
Pytho
n,
and
every
one of
them
is still
run-
ning
some-
where
right
now.
What

chang
ed
each
time
was
the
height
of the
lan-
guage
, not
wheth
er
some-
body
had

what should I call you?

this name stays in this browser. it is not
sent anywhere.

01

what
is
the
thin
g on
your
scree
n?

A page
is not a
place.
It is a

what should I call you?

this name stays in this browser. it is not
sent anywhere.

LOOKS.

The first
thing you
do with a
coding
agent is
ask it for a
website,
and a
minute
later there
is one.
The termi-
nal prints
an ad-

dress. You
open it
and the
thing you
described
is sitting
in the
browser
looking
finished.

Ask
where that
thing lives
and the
answer
comes
back as in

what should I call you?

this name stays in this browser. it is not
sent anywhere.

fine right
up until
something
breaks,
and then
none of
them tell
you where
to look.¹

¹
A book
note. A
model takes
the easiest
road. It

gave you localhost
with no backend,
because that is the
shortest way to
something that runs.

What arrives
when a page
opens?

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it called
index.html
with one
line inside.

```
<h1>hello
```

Open a terminal
in that folder and
run
python3 -m
http.server
8000. Go to local-

host:8000
in your
browser.
The word
hello is
there in
large let-
ters, arriv-
ing from a
server
small
enough
that noth-
ing can
hide inside
it.

Now

what should I call you?

this name stays in this browser. it is not
sent anywhere.

comes up
is the line
you typed,
character
for
character.

Change
the h1 to
a p and
refresh.

The same
word
comes
back
small.
Nobody
sent you a

smaller
word. You
changed
one in-
struction
and the
browser
drew the
letters dif-
ferently,
because
what trav-
elled to it
was never
a page. It
was a set
of instruc-

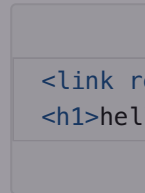
what should I call you?

this name stays in this browser. it is not
sent anywhere.

How does
the
browser
learn
about the
second
file?

One file is
not how
anything
real ar-
rives. Next
to in-
dex.html
make a file

called
style.css
containing
h1 { color:
crimson; }
and men-
tion it
from the
page.



Refresh
and the
word is
red.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

existed.
The only
file it had
been given
was in-
dex.html.
It read
that file
and came
to a line
mention-
ing a
stylesheet.
Only then
did it go
back to
the server

for a second file.
The page tells the browser what else to fetch, one discovery at a time.

Open the developer tools with Cmd and Option and I on a

what should I call you?

this name stays in this browser. it is not sent anywhere.

with the panel open. Two rows come up, index.html first and style.css second, and they can never come in the other order.

A slow page is rarely one

slow file.
The document arrives and mentions something.
That something arrives and mentions two more.
Nothing further down the chain starts un-

what should I call you?

this name stays in this browser. it is not sent anywhere.

does the page go when the server stops?

'The page is on my computer, so it will still be there,' you may be thinking.
In the Network

panel
there is a
checkbox
marked
Disable
cache.
Tick it so
the brows-
er stops
keeping
copies of
its own.
Now go to
the termi-
nal and
stop the
server

what should I call you?

this name stays in this browser. it is not
sent anywhere.

gone and
there is no
error
about
your
HTML,
because
your
HTML
never ar-
rived.
Nothing
had been
sitting in
the brows-
er waiting
for you.

Start
the server
again and
refresh.
Hello
comes
back in
red exact-
ly as be-
fore,
drawn
from noth-
ing. A
page is a
delivery
that hap-
pens again

what should I call you?

this name stays in this browser. it is not
sent anywhere.

look like
in a real
project?

Open the
project
you have
been
working
on and re-
fresh it
with the
panel
open. The
same list
appears,

only far
longer.

The document sits
at the top.
Everything the document
mentioned sits under
it, and under those
sits everything they
mentioned in turn.

At the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

much before they
see anything at
all, out of their
connection and their
phone plan.

That leaves one
question for the
rest of the book.
Some pro-

gram on
some ma-
chine sent
those files.
On the
hello page
you know
which one,
because
you start-
ed it your-
self and it
is still sit-
ting in
your ter-
minal. On
your real

what should I call you?

this name stays in this browser. it is not
sent anywhere.

CHE
CK

Op
en
the
Ne
tw
ork
pa
nel
on
the
pro

ject
t
yo
u
are
pro
ud-
est
of
an
d
re-
fre
sh
it.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

s
ar-
riv
ed?

Ho
w
ma
ny
kil
ob
yte
s
did
the

y
co
me
to?

W
hic
h
file
in
tha
t
list
can
yo

what should I call you?

this name stays in this browser. it is not
sent anywhere.

?
Th
ere
is
nea
rly
al-
wa
ys
one
.
Us
ual
ly
a

fon
t
yo
u
sto
pp
ed
us-
ing
mo
nth
s
ago
or
a

what should I call you?

this name stays in this browser. it is not
sent anywhere.

riv
ed
at-
tac
he
d
to
so
me
thi
ng
els
e.
No

thi
ng
an-
no
un
ced
it,
be-
cau
se
it
wa
s
me
nti

what should I call you?

this name stays in this browser. it is not
sent anywhere.

t
wa
s
it-
self
me
nti
one
d
by
a
file
.

No
thi
ng
nee
ds
re-
mo
vin
g
to-
da
y.
It
has
bee

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ry
visi
tor
sin
ce
the
da
y
it
ap-
pe
are
d.

☐ my
project
passes
this
check

You just
started a
server in-
side a
folder
without
my having
said what
a folder is,

what should I call you?

this name stays in this browser. it is not
sent anywhere.

files come
from. So
the termi-
nal and
the folder
come next.

why
can
your
de-
ploy
not
find
your
files?

The same
command

what should I call you?

this name stays in this browser. it is not
sent anywhere.

here.

You ran a
command
inside a
folder a
moment
ago and a
server ap-
peared,
and I nev-
er said
which
folder it
was read-

ing or why
that mat-
tered. The
same com-
mand run
from one
step high-
er up
would
have
served
nothing.

A
whole
family of
failures
comes out

what should I call you?

this name stays in this browser. it is not
sent anywhere.

not found
on the de-
ploy is
one, the
image that
loads lo-
cally and
404s in
production
is another.
In every
one of
them a
program is
standing
somewhere
you did

not expect, reading a path that means something different from there.

Where are you standing?

Open a

what should I call you?

this name stays in this browser. it is not sent anywhere.

one your development tools installed rather than the old black box.

A command never runs in general. It runs while you are standing somewhere and where

you are
standing
changes
what it
does.

pwd

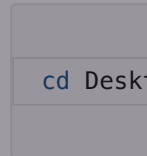
Print
working
directory.
It answers
with the
absolute
path of
where you
are stand-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

lists files
and fold-
ers togeth-
er with no
way to tell
them
apart, and
-F puts a
slash on
the end of
every fold-
er name.
Do not try
to work it
out from
the name.
README

has no dot
and is a
file, while
a folder
called
backup.old
has one
and is still
a folder.



Change di-
rectory,
into the
Desktop
that sits

what should I call you?

this name stays in this browser. it is not
sent anywhere.

has grown
by one
step. Run
ls -F and
the con-
tents have
changed,
because
the same
command
means
something
different
here.

```
cd ..
```

Two dots
is a real
entry that
exists in-
side every
folder on
the disk
and it
points at
that fold-
er's
parent.

That is
the whole

what should I call you?

this name stays in this browser. it is not
sent anywhere.

they work
the same
way on
every Mac
and every
Linux ma-
chine and
inside the
terminal
your
Windows
tools
installed.

Is the terminal a different place?

If the terminal still feels like a second machine hiding underneath the one you use, open Finder or Explorer

what should I call you?

this name stays in this browser. it is not sent anywhere.

One side of your screen and the terminal on the other.

Now cd your way to the same folder and run ls -F.

They are the same thing. The icons on

the left
and the
names on
the right
are one
folder de-
scribed in
two ways,
and the
three com-
mands you
just
learned
will reach
any folder
on the
disk.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

holds files

and other
folders

and those
hold more.

All of it
hangs

down from
one start-

ing point,
so your

whole disk
is a single

tree.

A path
is direc-
tions

through
that tree
and pwd
printed
one of
them.

```
/Users/c
```

It means
start at
the top of
the disk
and go
into Users,
then into
damla and

what should I call you?

this name stays in this browser. it is not
sent anywhere.

A postal
address is
written
the same
way and a
path is
that writ-
ten on one
line.

There
are two
kinds.

```
/Users/c  
style.cs
```

The absolute one begins with a slash and is measured from the top of the disk, so it means the same file wherever you use it. The relative one is measured

what should I call you?

this name stays in this browser. it is not sent anywhere.

different files depending on where they are read.

Why is this so hard to learn?

Go back to the folder with

index.html
and
style.css in
it and
start the
server
again. The
page loads
and the
word is
red. Now
stop the
server, run
cd .. to
step up
one level,
and start

what should I call you?

this name stays in this browser. it is not
sent anywhere.

Open lo-
calhost:80
00. There
is no page,
only a list
of folder
names.
Nothing
was moved
and noth-
ing was
deleted.
The href
inside your
HTML
still says
style.css

and it still
means the
style.css
next to
me, and
the server
is now
standing
one level
up where
no such
file sits.
Your
whole
project is
intact and
unreach-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

visible un-
til it costs
you a day.
A few
years ago
lecturers
in physics
and engi-
neering
started
running
into some-
thing they
could not
explain.
They
would ask

a class to
open a file
from a
particular
directory.
We as stu-
dents usu-
ally strug-
gle to do
exactly
that, my-
self includ-
ed, even
though we
use a com-
puter all
day and

what should I call you?

this name stays in this browser. it is not
sent anywhere.

you reach
it is to
search for
it.

That
model
works for
daily life.
Put every-
thing in
one pile
and search
when you
need
something.
You will
find your

holiday
pho-
tographs
every
time. It
stops
working at
the deploy,
because
the ma-
chine you
deploy to
has no
search
box. It is
handed a
path and

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ry struc-
ture is so
ordinary
to the
people
who know
it that
words for
it are hard
to find, so
nobody
explains it
and no-
body asks.

Why does
this de-
cide
whether
things
work?

Every
build step,
every de-
ploy tool
and every
error mes-
sage you
will read
for the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

File
does not
exist al-
most al-
ways
means a
relative
path was
read from
a folder
you were
not ex-
pecting.
The file
was there
all along,
and the

program
was stand-
ing some-
where else
when it
went look-
ing for it.
You saw
exactly
that a
minute
ago with a
server one
level too
high.

Someth
ing works

what should I call you?

this name stays in this browser. it is not
sent anywhere.

You typed
the com-
mand
while
standing
inside the
project
folder.
The other
machine
runs it
while
standing
wherever
it happens
to start, so
the rela-

tive path
points at a
place that
does not
exist.

Chapter
16 is
where this
bites
hardest.

MIT
teaches a
free course
for exactly
this gap
called The
Missing

what should I call you?

this name stays in this browser. it is not
sent anywhere.

comics
about the
shell that
are the
friendliest
thing on
the
subject.

CHE
CK

Op
en
a

terminal
and
leave
your
editor
closed

what should I call you?

this name stays in this browser. it is not
sent anywhere.

project
folder
using
only
pwd,
ls -
F
and

cd.

Re

ad

wh

at

co

me

s

bac

k

af-

ter

eac

h

ste

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ing

the

wh

ole

pat

h

fro

m

me

mo

ry.

W

he

n

yo
u
ar-
riv
e,
wh
ich
fol
der
in
tha
t
list
can
vo

what should I call you?

this name stays in this browser. it is not
sent anywhere.

con
ten
ts
of?
Th
ere
wil
l
be
one
,
usu
all
y

no
de
—
mo
dul
es
or
dis
t
or
.ne
xt.
Go
int

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ng-
ing
not
hin
g.
Th
e
ne
xt
tim
e
so
me
thi
ng

breaks that folder is going to appear in the

what should I call you?

this name stays in this browser. it is not sent anywhere.

which should not be the first time you see

the
na
me
).

☐ my
project
passes
this
check

Next
comes lo-
calhost,
and the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

friend.

why
does
your
link
only
open
for
you?

Your ma-
chine
answers
its own

what should I call you?

this name stays in this browser. it is not
sent anywhere.

finds
nothing.

This chap-
ter is
about fa-
mous lo-
calhost:30
00. Jokes
about it
have been
going
around the
internet
for a long
time.

Many people believe CS is dead because they built a website hosting on local. You click your own link and it opens. Your friend clicks it and it does not.

what should I call you?

this name stays in this browser. it is not sent anywhere.

What does localhost mean?

It begins with localhost:3000 or 127.0.0.1:3000, which are the same address writ-

ten two
ways. lo-
calhost is
a name
and
127.0.0.1
is the
number
the name
stands for.
RFC 6761
reserves
both for
one pur-
pose, and
that pur-
pose is to

what should I call you?

this name stays in this browser. it is not
sent anywhere.

chine
sends the
request to
a program
on itself.
The re-
quest nev-
er leaves
the ma-
chine,
which is
why this is
called the
loopback
address.
When
your

friend
types it,
their ma-
chine
sends the
request to
a program
on itself
and finds
none.

Each com-
puter's lo-
calhost is
a different
server.

The
number af-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

port is the
apartment
inside it.
One ma-
chine runs
many pro-
grams and
each waits
at its own
number.

3000 is a
convention
develop-
ment tools
picked,
nothing
more. The

port is
fine. The
building is
the prob-
lem, be-
cause you
gave your
friend the
name
every
building
uses for
itself.

Can your
own

what should I call you?

this name stays in this browser. it is not
sent anywhere.

into any
folder that
has a file
or two in
it. Run
this.

```
python3
```

This starts
a small
web
server.

The flag
tells it to
answer
only the

machine it
is running
on. Open
localhost:8
000 in the
browser
and you
see the
folder's
files listed.
Now type
the same
address
into your
phone's
browser.
Nothing

what should I call you?

this name stays in this browser. it is not
sent anywhere.

went look-
ing for a
program
at port
8000 on
itself.

Now
stop the
server
with
Ctrl+C
and start
it again
without
the flag.

python3

The flag
was doing
all the
work.
127.0.0.1
told the
server to
listen on
the loop-
back ad-
dress
alone, so
requests
arriving

what should I call you?

this name stays in this browser. it is not
sent anywhere.

0.0.0.0,
which is
not an ad-
dress you
connect
to. It
means
every net-
work in-
terface the
machine
has, in-
cluding
the wifi
card. A re-
quest from

the phone
is now ac-
cepted the
same as
one from
the ma-
chine it-
self. You
will see
0.0.0.0
again in
Docker
logs and in
cloud de-
ploy out-
put and it
always

what should I call you?

this name stays in this browser. it is not
sent anywhere.

the local
network.
On a Mac
that is ip-
config
getifaddr
en0 and
on Linux
hostname
-I. It looks
like
192.168.1.
24. If the
Mac com-
mand
comes
back emp-

ty your
wifi is on
another
interface,
so try en1.
Type
http://192
.168.1.24:8
000 into
the phone
and the
file list ap-
pears.
Nothing in
the files
changed,
only which

what should I call you?

this name stays in this browser. it is not
sent anywhere.

to look at.

How far
does the
local net-
work
reach?

'So I drop
the flag
and I am
live,' you
may be
thinking.
What you

have is a
server
reachable
from the
same wifi.
192.168.1.
24 is an
address in-
side your
home net-
work and
means
nothing
outside it,
so a friend
in a cafe
or a cus-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

closes,
nothing is
running
anywhere.

A prod-
uct has to
answer
while you
are asleep,
so the pro-
gram has
to run on
a machine
that stays
on and has
an address
the inter-

net can
route to.
Putting it
there is
called de-
ploying,
and that is
chapter
16. ir-
globe an-
swers at
three in
the morn-
ing be-
cause it is
not on my
laptop.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

at
the
ad-
dre-
ss
bar
of
the
pro-
jec-
t
yo-
u
are

pro
ud-
est
of.
If
it
be-
gin
s
wit
h
lo-
cal
hos
t.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

nu
mb
er
of
pe
opl
e
wh
o
can
op
en
it
is

1.
Th
en
clo
se
the
lap
top
.
Tu
rn
off
wif
i
on

what should I call you?

this name stays in this browser. it is not
sent anywhere.

u
are
on
mo
bil
e
dat
a
an
d
try
to
op
en

the
pro
jec
t.
W
hat
eve
r
co
me
s
bac
k
is
the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

er
it
ex-
ists
yet
.

☐ my
project
passes
this
check

After that
come the
two com-
mands you
run with-
out read-
ing them.
Only one
of npm
run dev
and npm
run build
makes a
thing you
can
deploy.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

what
is
the
dif-
fer-
ence
be-
twee
n
npm

what should I call you?

this name stays in this browser. it is not
sent anywhere.

run
build
?

One
builds
pages in
memory
for you.
The oth-
er
writes
the
folder
your
users
get.

Localhost
turned out
to be a
name your
machine
uses for it-
self. The
other half
of the ad-
dress is
still unex-
plained,
because
something
is answer-
ing at that
door, and

what should I call you?

this name stays in this browser. it is not
sent anywhere.

You
have run
that com-
mand a
thousand
times
without
wondering
what it is.
Sitting
next to it
in the
same file
there is
npm run
build. It
takes

much
longer and
produces a
folder you
have prob-
ably never
opened.
That fold-
er is the
part your
users get.

What is
running
when you

what should I call you?

this name stays in this browser. it is not
sent anywhere.

your ter-
minal and
waiting to
be asked
for things,
the same
way
python3 -
m
http.server
did in
chapter 1.
This one is
far clever-
er about

what it
hands
over.

Go to
the terminal where
it says
ready and
press Ctrl
and C.
Refresh
the browser.
The
site is
gone and
the error
is a re-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

door any
more.

Now
start it
again and
open the
project
folder,
looking for
the HTML
that arrived in
the browser.
There
is no such
file. Your
project

has a src
folder full
of compo-
nents and
not one of
them is a
page.

Nothing
on the
disk
matches
what you
saw in
view
source.

The
dev server

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it and
then let it
go. Its
whole job
is to read
the source,
work out
what the
browser
needs
right now
and hand
it over.
Then it
watches
the files so
it can do

the same
thing the
instant
you save.
Nothing
was writ-
ten down,
so a save
appears in
the brows-
er in un-
der a
second.

What
does the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

while,
where the
dev server
started in-
stantly.
When it
finishes
there is a
new folder
called dist
on Vite
and .next
on Next.js.
Go in
there with
the com-

mands
from chap-
ter 2.

```
cd dist  
ls -F
```

The
HTML
that did
not exist a
minute
ago is in
there. So
are the
scripts,
with

what should I call you?

this name stays in this browser. it is not
sent anywhere.

there so a
browser
can safely
keep an
old copy
without
mistaking
it for the
new one
after you
deploy.
The CSS
from every
component
in the
project
has been

gathered
into one
file.

Your
source is
written in
whatever
shape lets
you work.
This folder
is written
for
browsers,
in one
shape and
once. The
dev server

what should I call you?

this name stays in this browser. it is not
sent anywhere.

of it to
disk.

Which
one gets
deployed?

The folder
gets de-
ployed,
never your
source and
never the
dev server.
Deploying

means
building
that folder
and
putting its
contents
somewhere
that an-
swers re-
quests all
day, so a
visitor is
handed a
finished
file instead
of waiting
for a pro-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

right now.

```
cd dist  
python3
```

Open lo-
calhost:80
00. That is
your site,
coming
out of a
plain file
server that
has never
heard of
your
frame-

work. It is
the same
server that
handed
you one
line of
HTML in
chapter 1.

Why do
the two
behave
different-
ly?

what should I call you?

this name stays in this browser. it is not
sent anywhere.

be think-
ing. Open
one of
those
scripts in
dist and
read it.
Your func-
tion names
are gone,
replaced
by single
letters.
The spaces
are gone.

The whole
file is one
line.

The
dev server
is opti-
mised for
you. It
starts in-
stantly
and re-
freshes in-
stantly,
keeps your
variable
names so
errors stay

what should I call you?

this name stays in this browser. it is not
sent anywhere.

the visitor,
so it
makes the
smallest
files it can,
squashes
everything
together,
drops code
that noth-
ing uses
and short-
ens every
name it
can reach.

Squashi
ng changes
behaviour.
Code that
quietly re-
lied on a
function
still hav-
ing its
own name
breaks, be-
cause the
name is
gone and
you just
read the
file where

what should I call you?

this name stays in this browser. it is not
sent anywhere.

into the
file, so
changing
it on the
server af-
terwards
changes
nothing
until you
build
again, and
chapter 13
is about
the trou-
ble that
causes. A
file the

dev server
found by
walking
your folder
is missing
from the
output en-
tirely if
nothing
imported
it
properly.

So a
project
can run
beautifully
for weeks

what should I call you?

this name stays in this browser. it is not
sent anywhere.

having
changed.
When
somebody
says it
works on
my ma-
chine they
are usually
telling the
truth, be-
cause their
machine
was run-
ning the
friendly
version.

Before
you de-
ploy, run
the build
and serve
the output
locally the
way you
just did,
then click
around
your own
site.
Anything
broken in
that folder
is broken

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it.

CHE
CK

Ru
n
np
m
run
bui
ld
on
yo

ur
ow
n
pro
jec
t
an
d
let
it
fin-
ish.

Ho
w

what should I call you?

this name stays in this browser. it is not
sent anywhere.

W
hat
eve
r
tha
t
nu
mb
er
is,
it
ha
pp
ens
aga

in
on
eve
ry
de-
plo
y.

W
hat
is
in-
sid
e
the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

h
cd
dis
t
or
cd
.ne
xt
an
d
run
ls -
F,
the
n

find
the
HTML
your
source
never
contain
ne

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it
with
python3
-m
http
server
8000
and

use
yo
ur
ow
n
site
co
mi
ng
out
of
a
file
ser
ver

what should I call you?

this name stays in this browser. it is not
sent anywhere.

hin
g
ab
out
yo
ur
fra
me
wo
rk.
If
so
me
thi

ng
wo
rks
un-
der
np
m
run
de
v
an
d
not
in
the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

nd
a
rea
l
bu
g
mo
nth
s
be-
for
e a
str
an

ger
wo
uld
ha
ve
fou
nd
it
for
yo
u.

☐ my
project
passes
this

what should I call you?

this name stays in this browser. it is not
sent anywhere.

front and
the till.
What a
backend
is, and
why some
things can
never live
in the files
you just
looked at
however
well you
hide them.

why
does
the
till
stan
d in
a
dif-
fer-
ent

what should I call you?

this name stays in this browser. it is not
sent anywhere.

browser
belongs
to who-
ever is
standing
there.

Everythin
g so far
has hap-
pened in
one room.
Files ar-
rive and
the brows-
er draws

them. You
have now
read your
own
project
the way a
visitor
reads it. A
shop has a
second
room be-
hind the
counter
where the
till and
the keys
are, and

what should I call you?

this name stays in this browser. it is not
sent anywhere.

look like
folders sit-
ting next
to each
other in
the same
editor, so
the till
ends up
out on the
shop floor.

Which
room is

the
browser?

The
browser is
the front
one and
everything
in it be-
longs to
whoever is
standing
there.

Every
file that
arrived is
listed in

what should I call you?

this name stays in this browser. it is not
sent anywhere.

it, includ-
ing the
scripts you
wrote.

That will
not be
patched
one day,
because it
is what
delivery
means.

'My
code is
minified,
so nobody

can read
it,' you
may be
thinking.
Open one
of your
own
scripts in
that panel
and find
the button
marked
with
braces at
the bot-
tom of the
view. The

what should I call you?

this name stays in this browser. it is not
sent anywhere.

makes the
file smaller
and it
hides
nothing.

What is a
backend?

A backend
is a pro-
gram on
another
machine
that an-

swers re-
quests. It
runs some-
where else
and listens
at a door.
You talk
to it by
sending
exactly
the kind of
request
you
watched
arrive in
chapter 1.

The

what should I call you?

this name stays in this browser. it is not
sent anywhere.

come into
the
kitchen,
because
the
kitchen is
a different
room. A
script run-
ning in the
browser is
a locked
cupboard
standing
in the din-
ing room,
and the

customer
has all
evening.

What is
an API?

An API is
the menu.
A menu
tells you
what you
may order
and in
what
form. An

what should I call you?

this name stays in this browser. it is not
sent anywhere.

meaning
this ad-
dress with
this
method
and these
fields.

Anything
not on the
menu is
refused.

Writing
one is the
act of de-
ciding
what

strangers
may ask
you for.

The
verbs on
that menu
are the
methods.
GET
fetches
something,
POST
sends
something
new, PUT
replaces,
DELETE

what should I call you?

this name stays in this browser. it is not
sent anywhere.

and search
crawlers
act on
that un-
derstand-
ing. That
is how a
link which
deletes a
row gets
triggered
by a
crawler
that was
only look-
ing
around.

What do
the num-
bers
mean?

Every an-
swer
comes
back with
a status
number
and MDN
carries the
definitions
if you
want them
properly

what should I call you?

this name stays in this browser. it is not
sent anywhere.

under
stood you
perfectly
and has no
such dish.
500 means
the
kitchen
broke
while
cooking,
so the
fault is on
their side.

The
two that
get con-

fused constantly are
401 and
403. A 401
means the
kitchen
does not
know who
you are,
while a
403 means
it knows
exactly
who you
are and re-
fuses any-
way.

what should I call you?

this name stays in this browser. it is not
sent anywhere.

fixes, and
telling
them
apart is
the whole
of chapter
26.

Can you
break

your own
rule?

Find
something
in your
own
project
that the
page refus-
es to let
you do. A
submit
button
that stays
disabled
until the

what should I call you?

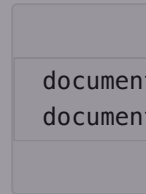
this name stays in this browser. it is not
sent anywhere.

that only
appears
when you
are logged
in.

Anything
the page
decides.

Open
the devel-
oper tools
and go to
the
Console
tab. Then
type one

of these,
matching
whichever
rule you
found.



The but-
ton is live.
The field
takes as
much as
you want.
You did
not find a

what should I call you?

this name stays in this browser. it is not
sent anywhere.

the same
console
you just
used.

That is
why a
check
written
into your
page is a
suggestion.
Your
script runs
on their
machine.
A check
that holds

has to run
in a room
they can-
not reach,
which is
the room
this chap-
ter is
about.

Where
does the
key go?

Into the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

sends a
photo-
graph to a
vision
model and
that mod-
el charges
per call.
The key
that pays
for it be-
longs in
the till, so
the brows-
er never
holds it.
The pho-

tograph
goes to a
Cloudflare
Worker,
the
Worker
holds the
key and
calls the
model,
and the
answer
comes
back.
Anyone
can open
the

what should I call you?

this name stays in this browser. it is not
sent anywhere.

their ma-
chine and
find noth-
ing in
them to
take.

The
wrong ver-
sion is one
line of dif-
ference.
The
browser
calls the
model di-
rectly with
the key at-

tached to
the re-
quest. It
works on
the first
try (which
is the
trouble)
because
from be-
hind the
counter
both ver-
sions look
the same
to you.
Every visi-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

is about
the size of
that bill.

CHE
CK

Op
en
yo
ur
ow
n
pro

ject
with
the
Network
panel
opened
and
used

what should I call you?

this name stays in this browser. it is not
sent anywhere.

start
appearing
. Where
each
re-

qu
est
go-
ing
?
Re
ad
the
ad-
dre
ss
of
eac
h
one

what should I call you?

this name stays in this browser. it is not
sent anywhere.

ht
to
so
me
bo
dy
els
e's
ser
vic
e
the
n
yo

ur
bro
ws
er
is
tal
kin
g
to
tha
t
ser
vic
e
di-

what should I call you?

this name stays in this browser. it is not
sent anywhere.

thi
ng
in
tha
t
re-
qu
est
is
pu
bli
c.

What
in
your
code
costs
money
or
grants

what should I call you?

this name stays in this browser. it is not
sent anywhere.

key
, a
database
URL,
an
admin
token
.

W
her
e
do
es
it
live
rig
ht
no
w?

If
it
rea

what should I call you?

this name stays in this browser. it is not
sent anywhere.

the
n
it
is
pu
bli
c,
an
d
the
ne
xt
tw
o
cha

pte
rs
are
ab
out
mo
vin
g
it.

☐ my
project
passes
this
check

what should I call you?

this name stays in this browser. it is not
sent anywhere.

file works
perfectly
until two
people use
your
project at
the same
time.